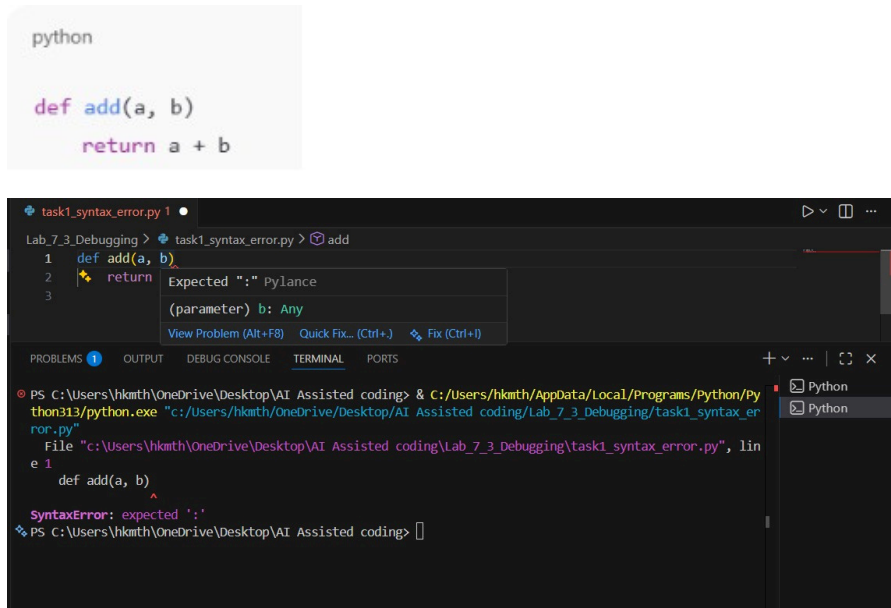


AI Assisted Coding Assignment - 7

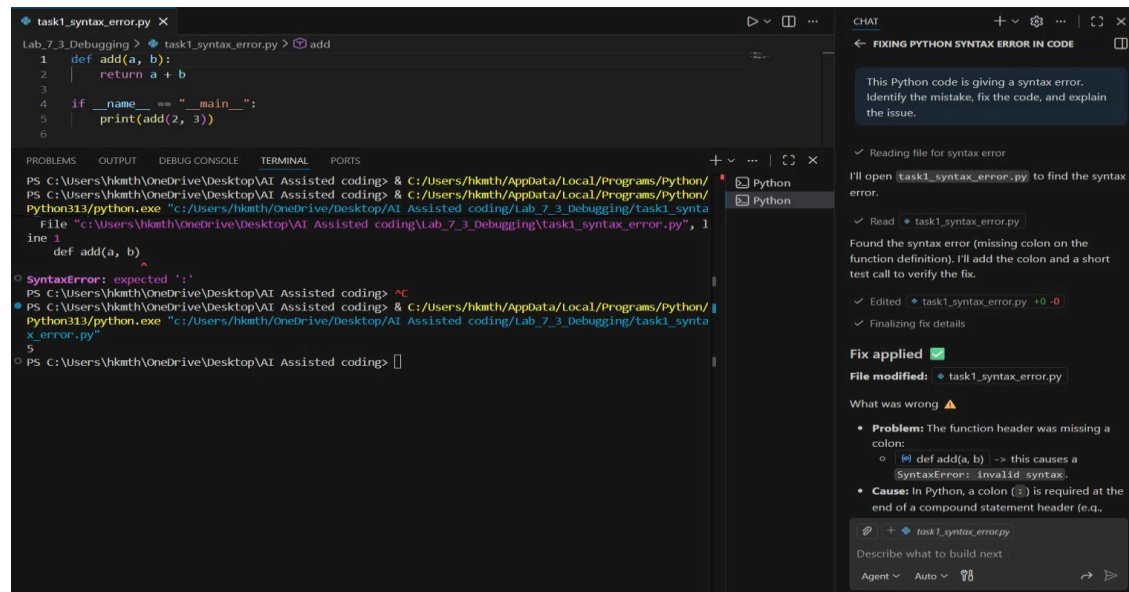
Chilukala Sai Architha(2303A52220) - Batch 37

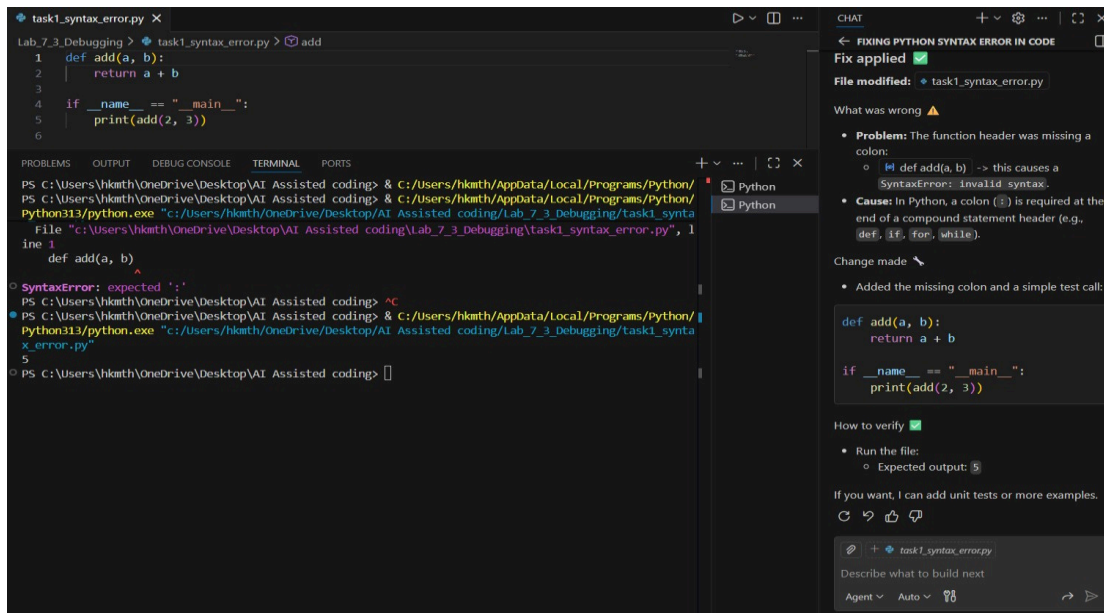
Task 1: Fixing Syntax Errors

You are reviewing a Python program where a basic function definition contains a syntax error.



Prompt: This Python code is giving a syntax error. Identify the mistake, fix the code, and explain the issue.





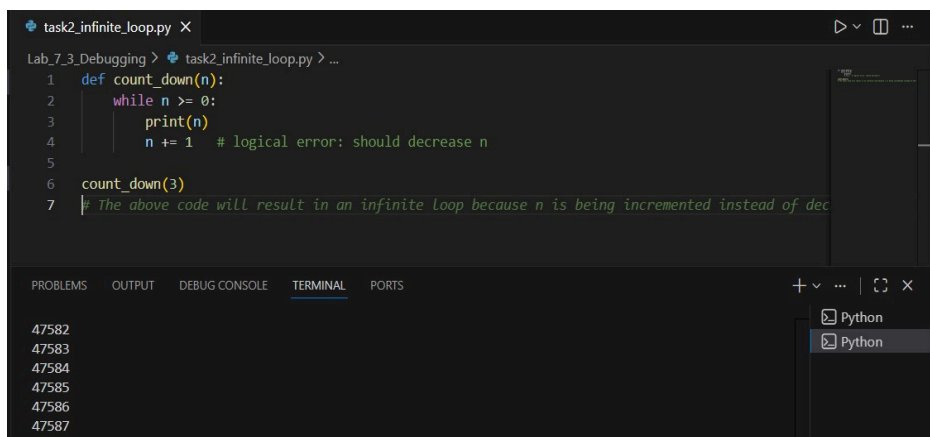
Explanation: The buggy Python code is executed in VS Code to observe the syntax error. The error is then fixed using AI suggestions, and the corrected code runs successfully.

Task 2: Debugging Logic Errors in Loops

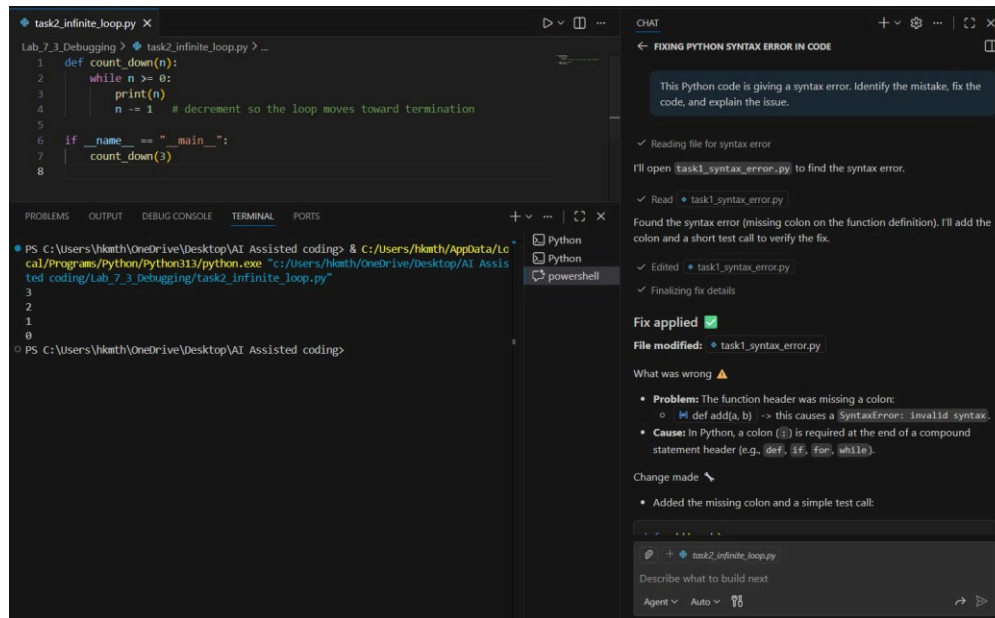
You are debugging a loop that runs infinitely due to a logical mistake.

```
python

def count_down(n):
    while n >= 0:
        print(n)
        n += 1 # Should be n -= 1
```



Prompt : The following Python function runs infinitely due to a logic error in the loop. Identify the mistake, explain why the loop does not terminate, and correct the code so that the countdown works properly.



The screenshot shows a VS Code editor with a file named `task2_infinite_loop.py`. The code is as follows:

```
1 def count_down(n):
2     while n >= 0:
3         print(n)
4         n -= 1 # decrement so the loop moves toward termination
5
6 if __name__ == "__main__":
7     count_down(3)
8
```

The chat window on the right, titled "FIXING PYTHON SYNTAX ERROR IN CODE", contains the following text:

This Python code is giving a syntax error. Identify the mistake, fix the code, and explain the issue.

- ✓ Reading file for syntax error
- I'll open `task1_syntax_error.py` to find the syntax error.
- ✓ Read `task1_syntax_error.py`
- Found the syntax error (missing colon on the function definition). I'll add the colon and a short test call to verify the fix.
- ✓ Edited `task1_syntax_error.py`
- ✓ Finalizing fix details

Fix applied ✓

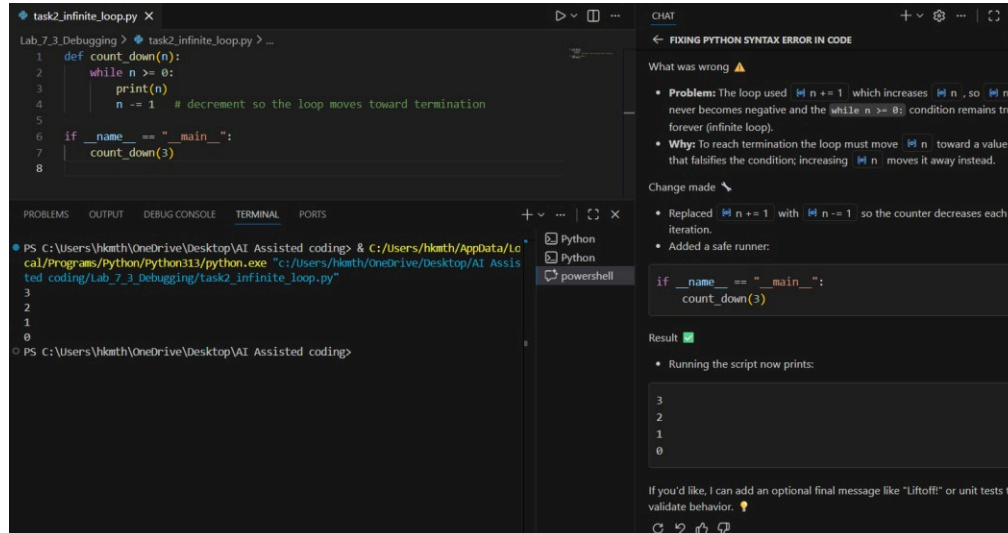
File modified: `task1_syntax_error.py`

What was wrong ⚠

- **Problem:** The function header was missing a colon:
 - `def add(a, b)` -> this causes a `SyntaxError: invalid syntax`.
- **Cause:** In Python, a colon (`:`) is required at the end of a compound statement header (e.g., `def`, `if`, `for`, `while`).

Change made ↗

- Added the missing colon and a simple test call:



The screenshot shows the same VS Code editor with the file `task2_infinite_loop.py`. The chat window on the right, titled "FIXING PYTHON SYNTAX ERROR IN CODE", contains the following text:

What was wrong ⚠

- **Problem:** The loop used `n += 1` which increases `n`, so `n` never becomes negative and the `while n >= 0:` condition remains true forever (infinite loop).
- **Why:** To reach termination the loop must move `n` toward a value that falsifies the condition; increasing `n` moves it away instead.

Change made ↗

- Replaced `n += 1` with `n -= 1` so the counter decreases each iteration.
- Added a safe runner:

```
if __name__ == "__main__":
    count_down(3)
```

Result ✓

- Running the script now prints:

```
3
2
1
0
```

If you'd like, I can add an optional final message like "Liftoff!" or unit tests to validate behavior. 🌟

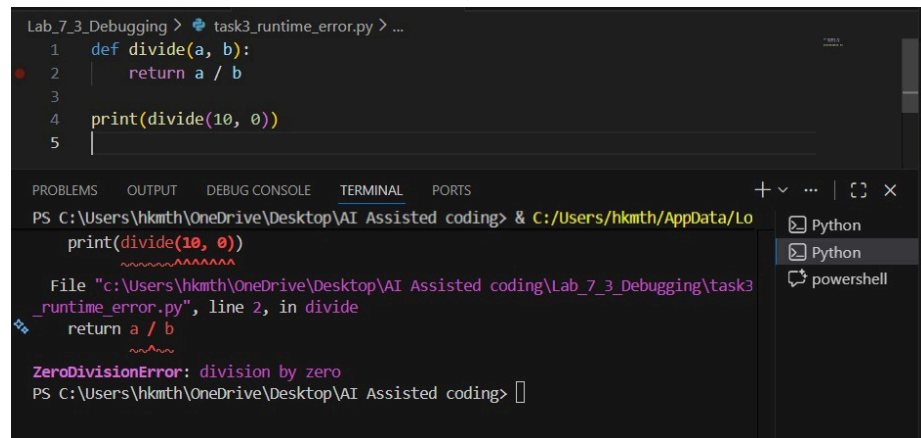
Explanation: The loop runs infinitely because the value of `n` is increased instead of decreased, so the loop condition always remains true. The AI identified this logical error and corrected the update statement. After fixing it, the loop properly counts down and terminates.

Task 3: Handling Runtime Errors (Division by Zero)

A Python function crashes during execution due to a division by zero error.

```
# Debug the following code
def divide(a, b):
    return a / b

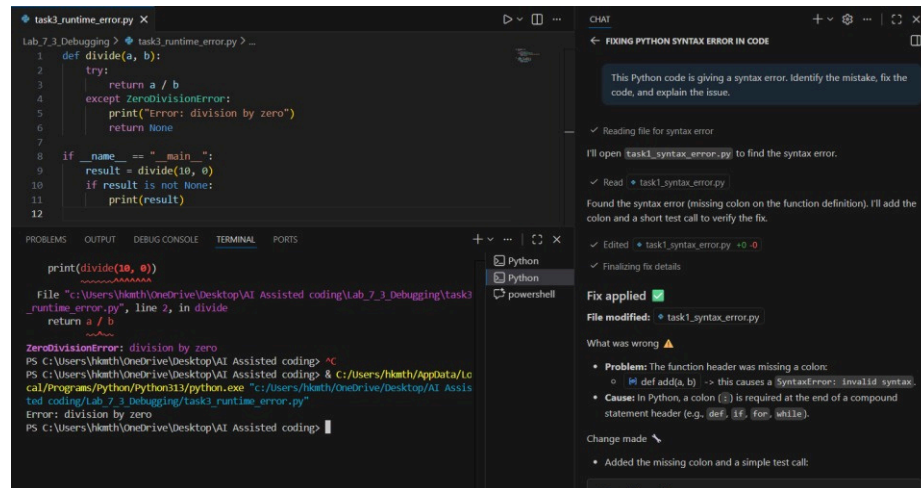
print(divide(10, 0))
```



```
Lab_7_3_Debugging > task3_runtime_error.py > ...
1 def divide(a, b):
2     return a / b
3
4 print(divide(10, 0))
5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hkmth\OneDrive\Desktop\AI Assisted coding> & C:/Users/hkmth/AppData/Lo
print(divide(10, 0))
~~~~~
File "c:\Users\hkmth\OneDrive\Desktop\AI Assisted coding\Lab_7_3_Debugging\task3
_runtime_error.py", line 2, in divide
    return a / b
~~~~~
ZeroDivisionError: division by zero
PS C:\Users\hkmth\OneDrive\Desktop\AI Assisted coding>
```

Prompt: The following Python function causes a runtime error during execution. Identify the error, explain why it occurs, and modify the code using try-except to handle the error safely.



```
task3_runtime_error.py X
Lab_7_3_Debugging > task3_runtime_error.py > ...
1 def divide(a, b):
2     try:
3         return a / b
4     except ZeroDivisionError:
5         print("Error: division by zero")
6         return None
7
8 if __name__ == "__main__":
9     result = divide(10, 0)
10    if result is not None:
11        print(result)
12
```

```
print(divide(10, 0))
~~~~~
File "c:\Users\hkmth\OneDrive\Desktop\AI Assisted coding\Lab_7_3_Debugging\task3
_runtime_error.py", line 2, in divide
    return a / b
~~~~~
ZeroDivisionError: division by zero
PS C:\Users\hkmth\OneDrive\Desktop\AI Assisted coding> ^C
PS C:\Users\hkmth\OneDrive\Desktop\AI Assisted coding> & C:/Users/hkmth/AppData/Lo
cal/Programs/Python/python313/python.exe "c:\Users\hkmth\OneDrive\Desktop\AI Assis
ted coding\Lab_7_3_Debugging\task3_runtime_error.py"
Error: division by zero
PS C:\Users\hkmth\OneDrive\Desktop\AI Assisted coding>
```

CHAT

← FIXING PYTHON SYNTAX ERROR IN CODE

This Python code is giving a syntax error. Identify the mistake, fix the code, and explain the issue.

- ✓ Reading file for syntax error
- I'll open `task1_syntax_error.py` to find the syntax error.
- ✓ Read `task1_syntax_error.py`
- Found the syntax error (missing colon on the function definition). I'll add the colon and a short test call to verify the fix.
- ✓ Edited `task1_syntax_error.py` +5 -0
- ✓ Finalizing fix details

Fix applied ☒

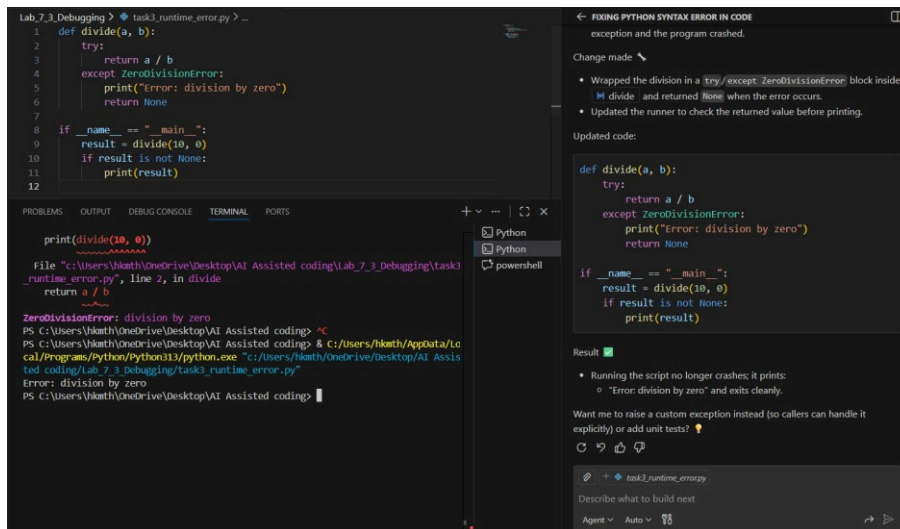
File modified: `task1_syntax_error.py`

What was wrong

- **Problem:** The function header was missing a colon:
 - o `def add(a, b)` -> this causes a `SyntaxError: invalid syntax`.
- **Cause:** In Python, a colon (`:`) is required at the end of a compound statement header (e.g., `def`, `if`, `for`, `while`).

Change made

- Added the missing colon and a simple test call:



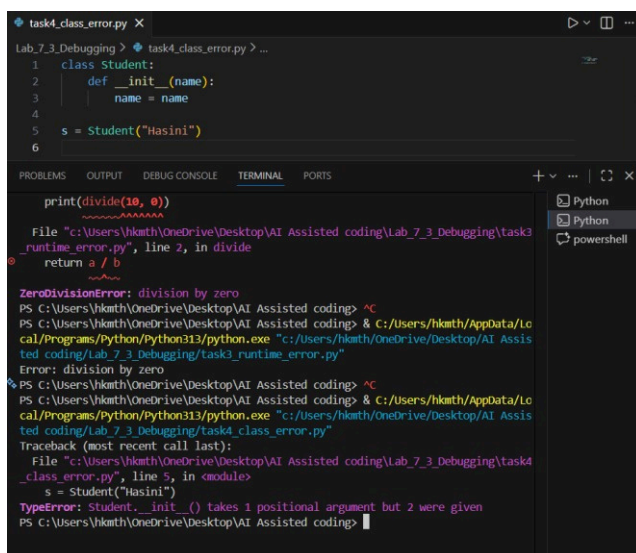
Explanation: The program crashes at runtime because it attempts to divide a number by zero. This results in a Zero Division Error, which is a runtime error.

Task 4: Debugging Class Definition Errors

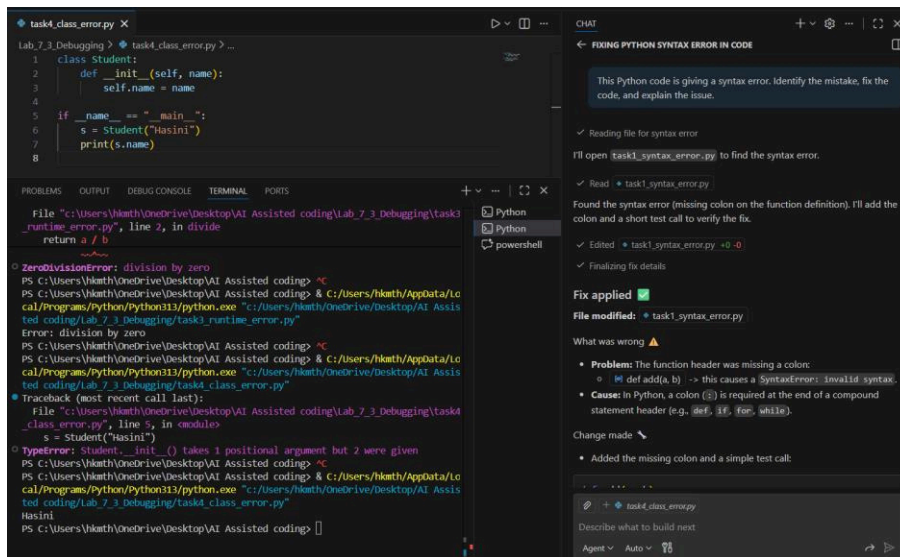
You are given a faulty Python class where the constructor is incorrectly defined.

```
python

class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```



Prompt: The following Python class is giving an error during object creation. Identify the issue in the constructor, correct the class definition, and explain why the error occurs.



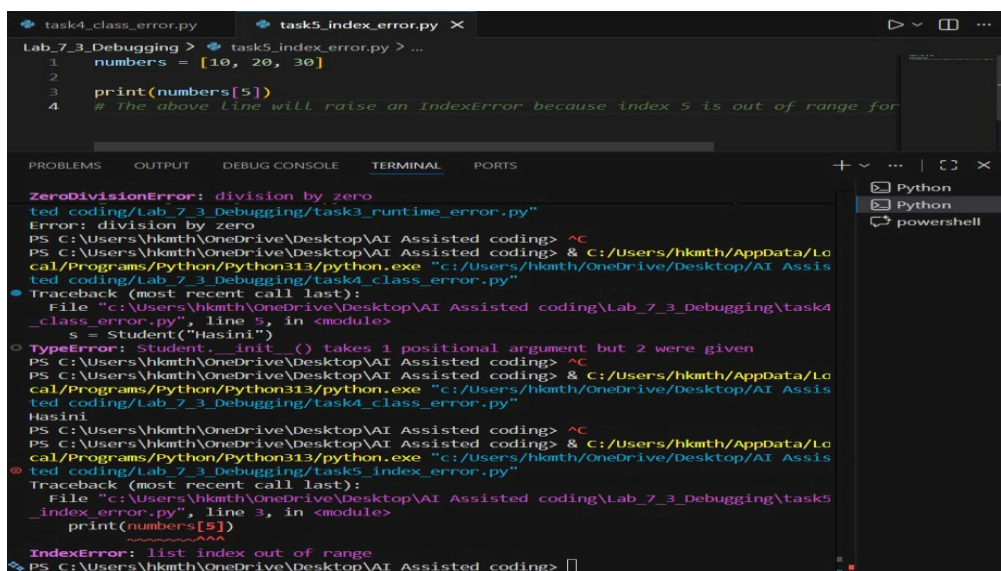
Explanation: In this task, a runtime error occurs because the `self` parameter is missing in the class constructor. Python automatically passes the object reference to methods, and without `self`, the arguments do not match. The AI fixes the issue by adding `self` to the constructor definition.

Task 5: Resolving Index Errors in Lists

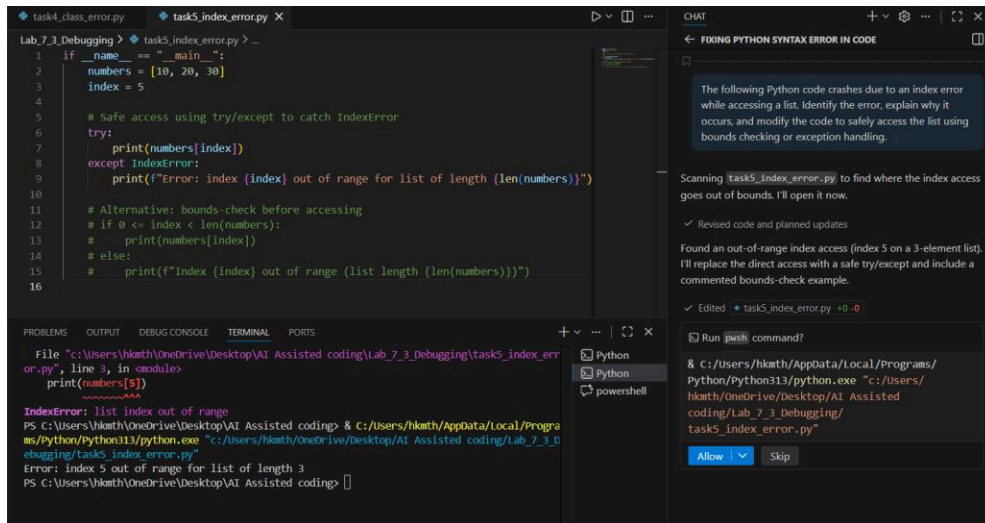
A program crashes when accessing an invalid index in a list.

```
python

numbers = [1, 2, 3]
print(numbers[5])
```



Prompt: The following Python code crashes due to an index error while accessing a list. Identify the error, explain why it occurs, and modify the code to safely access the list using bounds checking or exception handling.



The screenshot shows a VS Code editor with two tabs: `task4_class_error.py` and `task5_index_error.py`. The active tab is `task5_index_error.py`, which contains the following Python code:

```
1 if __name__ == "__main__":
2     numbers = [10, 20, 30]
3     index = 5
4
5     # Safe access using try/except to catch IndexError
6     try:
7         print(numbers[index])
8     except IndexError:
9         print(f"Error: index {index} out of range for list of length {len(numbers)}")
10
11    # Alternative: bounds-check before accessing
12    # if 0 <= index < len(numbers):
13    #     print(numbers[index])
14    # else:
15    #     print(f"Index {index} out of range (list length {len(numbers)})")
16
```

The chat window on the right, titled "FIXING PYTHON SYNTAX ERROR IN CODE", contains the following text:

The following Python code crashes due to an index error while accessing a list. Identify the error, explain why it occurs, and modify the code to safely access the list using bounds checking or exception handling.

Scanning `task5_index_error.py` to find where the index access goes out of bounds. I'll open it now.

✓ Revised code and planned updates

Found an out-of-range index access (index 5 on a 3-element list). I'll replace the direct access with a safe try/except and include a commented bounds-check example.

✓ Edited `task5_index_error.py` +0 -0

Run `python` command?

& C:/Users/hkath/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/hkath/OneDrive/Desktop/AI Assisted coding/Lab_7_3_Debugging/task5_index_error.py"

Allow Skip

The terminal at the bottom shows the execution of the code, which results in an `IndexError: list index out of range` at line 3, in `module`, `print(numbers[5])`. The error message also indicates the list length is 3.

Explanation: In this task, the program crashes because it tries to access a list index that does not exist. This causes an Index Error at runtime. The AI fixes the issue by handling the error safely, preventing the program from crashing.